Venkata Lakshmi Parimala Pasupuleti

Business Intelligence · Semantic Modeling · Data Analytics Engineering · AI-driven BI · Power BI · Python · SQL · Data Governance

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PROFESSIONAL SUMMARY

Data analytics engineer with 8+ years of production data engineering at scale and a M.S. in Data Analytics Engineering (GMU, GPA 3.87). Built and optimized enterprise data pipelines, reporting systems, and AI-powered analytics for 100K+ monthly active users at Entrata — translating complex compliance and financial data into actionable insights for stakeholders across Product, Operations, Legal, and Finance. Experienced in end-to-end analytics delivery: data ingestion, transformation, modeling, visualization (Power BI, Tableau, ggplot2), and governance. Applied predictive modeling and AI-driven analytics to accelerate research and decision-making at GMU. Proven cross-functional communicator who translates data complexity into business-ready insights and self-service solutions.

CORE COMPETENCIES

Business Intelligence & Reporting | Semantic & Dimensional Modeling | Power BI & Tableau Dashboards | AI-driven BI & Analytics | Data Pipeline & ETL Engineering | Data Governance & Compliance | Python & SQL at Scale | Statistical Modeling & ML | Cross-Functional Stakeholder Delivery | Self-Service Analytics Enablement | Observability & Data Quality | Predictive Insights

EXPERIENCE

George Mason University — Research Associate

Aug 2024 – Aug 2025

Data Analytics · Statistical Modeling · AI-Powered Insights · BI Dashboards

- Built **multi-source prediction models** (Python, R) achieving $R^2 = 0.977$ across 190+ countries on WHO and UK Road Safety datasets — data collection, transformation, hypothesis testing, and validation pipeline delivering research strategy recommendations to faculty stakeholders.
- Surfaced data integrity anomalies across **3 research pipelines** by applying Isolation Forest automation — replaced manual review with scalable, repeatable anomaly detection that improved downstream model reliability.
- **Improved research comprehension by 40%** across two cohorts by designing leadership analytics dashboards in **Tableau, ggplot2, and Quarto** — translating complex multi-country datasets into decision-ready visual narratives for faculty and administrative stakeholders.

George Mason University — Graduate Teaching Assistant

Aug 2025 – May 2026 · Part-time

Python Programming · Database Design · Data Visualization Instruction

- Mentored **600+ students** across Python programming (data structures, OOP, automation) and database design (ER modeling, SQL, normalization, pivot tables, data validation) — developed supplementary exercises and walkthroughs; graded with rubric-aligned feedback.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer — Data Engineering · Analytics · BI Reporting · Data Governance at Scale

- **Delivered enterprise analytics and reporting** across financial services workflows: HAP allocation, AMI-based pricing tiers, tax credit calculations, HUD payments, income limits — built data models, batch pipelines, and reporting solutions consumed by 100K+ monthly active users; stakeholders across Finance, Compliance, Operations, and Product.
- **Optimized 200+ SQL queries** across all client environments (30% performance gain) through execution plan tuning, indexing, database partitioning, caching strategies, views, stored procedures, and observability tooling — designed high-performance, ACID-compliant data access patterns for complex production schemas.
- Built and shipped **20+ full-stack analytics and reporting modules** (Python, React, PHP, PostgreSQL, RabbitMQ) — 99.5% uptime across 6+ months; led sprint planning across 120+ Agile sprints; collaborated cross-functionally with Product, UX, Compliance, and Legal to translate business requirements into scalable data solutions.
- **Data governance and security at scale**: 100% PHI compliance on AWS S3 migration (Libsodium-encrypted; zero security incidents); enforced access controls, data minimization, and audit logging across all resident PII; zero compliance violations across federal housing programs (LIHTC, HUD, HAP, AMI, Section 8).
- Implemented **Salesforce reporting automations, data sync, and CRM data pipelines** — connected product data to commercial reporting systems using Clay and n8n; automated workflows reducing manual reporting overhead.
- Eliminated 80% of manual operational workload via Python/Bash automation pipelines; mentored 5+ engineers; conducted 300+ peer code reviews; led microservices architecture redesign (Docker, 50% scalability improvement).

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Data Pipelines · Reporting · Compliance Data

- Built data pipelines processing **1.6M HUD/Tax Credit income limit records** across five years; designed generic SQL scripts for data import and eligibility modeling; created 56 state-specific compliance reporting documents; migrated client data to AWS S3.

ANALYTICS PROJECTS

CrashClarity — Large-Scale Predictive Analytics (104K Records)

ML Analytics · BI Dashboard · Predictive Insights

2025 · Python, R, scikit-learn, Random Forest, Logistic Regression, Feature Engineering, GitHub Pages · View Project

- Achieved **85%+ accuracy on 104K+ UK accident records** by building a Random Forest + Logistic Regression pipeline with feature engineering across weather, road, and time variables — end-to-end: data ingestion, transformation, model training, validation, and self-service interactive dashboard deployed on GitHub Pages.
- Translated complex multi-variable accident patterns into **decision-ready visualizations** with clear business narratives — same storytelling approach required for translating Partner and Marketing data into actionable insights.

AI on FHIR — NLP-Driven Data Query & Analytics Platform

AI-Powered Analytics · Natural Language Query · Governed Data Access

2024 · Flask, LLMs, FHIR, React, Docker, spaCy, BERT, Embeddings · github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced manual clinical data retrieval time ~60% by building a Copilot-style natural language query platform over structured medical records — LLMs + BERT translate plain-language questions (EN/ES) into FHIR API queries against a governed semantic modeling layer; HIPAA-compliant RBAC enforced; analogous to AI-driven BI and natural language querying over Power BI semantic models.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Analytics & BI: Python, R, SQL, PL/SQL, Power BI, Tableau, ggplot2, Quarto, Data Visualization, Statistical Modeling, Feature Engineering, Predictive Analytics, Anomaly Detection, Isolation Forest, Random Forest, Logistic Regression, XGBoost, Scikit-learn, PyTorch, NumPy, Pandas, Jupyter

Data Engineering & Governance: Data Pipelines, ETL, Data Architecture, Semantic Modeling, Dimensional Modeling, Star Schema, Data Modeling, Batch Pipelines, Data Governance, Data Quality, Observability, Access Controls, Row-Level Security, PII Handling, PHI Compliance, Audit Logging, Data Minimization, AWS (S3, Glue, EC2, RDS), PostgreSQL, MySQL, Oracle, RabbitMQ

AI-Powered Analytics: AI-driven BI, AI-Driven Insights, Copilot-Style NLQ, Natural Language Querying, LLM Integration, RAG Pipelines, Embeddings, Semantic Search, NLP, spaCy, BERT, Hugging Face, GenAI, Anthropic API, OpenAI API, LangChain, LangGraph, FastAPI

Platform & Delivery: React, JavaScript, TypeScript, Docker, CI/CD (GitHub Actions), Microservices, Salesforce (reporting automations, data sync, CRM pipelines), Clay, n8n, Git, Linux, Agile/Scrum, Cross-Functional Delivery